The Past, Present, and Future of Performance Engineering

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Alex Podelko

- Specializes in performance since 1997
- Senior Performance Engineer at AWS – Amazon Aurora
  - Before worked for MongoDB, Oracle/Hyperion, Intel, and Aetna
- SPEC RG Steering Committee Member



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Agenda

- The Past: A Short History of Performance Engineering
- The Present
  - Performance Gets More Attention
  - Adjusting Performance Engineering to Major Industry Trends
  - Performance Engineering as an Independent Discipline
  - AI Current Impact
- The Future
  - AI Coming Impact – Paradigm Change ?

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Computing Paradigms

Looks like a new wave of performance engineering starts 10+ years after new computing paradigm

Mainframes - Late 50s

Distributed Systems [Decentralization] - Late 70s

Web [Centralization] - Late 90s

Cloud, Mobile [Further centralization] - Late 00s

~~? - Peer-to-Peer, IoT, Fog, Edge [Decentralization] - Late 10s~~

AI – Late 10s

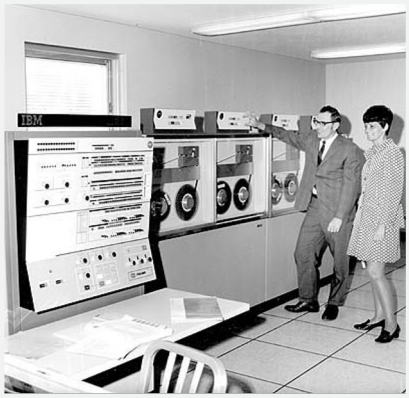
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1966: Instrumentation

1966 – SMF (System Management Facilities) released as part of OS/360

- Still in use

Observability
IT Operations
Analytics
Big Data
Deep Diagnostics



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1968: Response Times

1968 - Robert Miller (IBM) in his "Response Time in Man-Computer Conversational Transactions" paper described several threshold levels of human attention

- Widely cited by many later researchers



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1974: Monitoring

1974 – RMF (Resource Measurement Facility) released as part of MVS

- Still in use

1975 – OMEGAMON for MVS by Candle

- Claimed to be the first real-time monitor
- Acquired by IBM in 2004



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1975: Community

1975 – The Computer Measurement Group (CMG) was created, holding annual conferences until recently

CMG Computer Measurement Group

http://cmg.org

CMG
YOUR TECHNOLOGY COMMUNITY

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1977: Performance Analysis Tool

1977 – BEST/1 was released by BGS Systems, capacity and performance management tool

- the first commercial package for computer performance analysis to be based on analytic models.

BGS Systems was acquired by BMC Software in 1998

```
                          Work with BEST/1 Model

Performance data . . . :   BESTDATA  (Q930400309)
Model/Text . . . . . . :   BEST1MODEL

Select one of the following:

     1. Work with workloads
     2. Specify objectives and active jobs

     5. Analyze current model
     6. Analyze current model and give recommendations
     7. Specify workload growth and analyze model

    10. Configuration menu
    11. Work with results

                                                                    More ...
```

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Late 70s: Distributed Systems

Distributed Systems

- Client/Server, Three-Tier, N-Tier, etc.

Less control, less instrumentation

More need for performance design and testing



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1987: Load Testing Tools

1987 – PreVue by Performance Awareness

- Acquired by Rational Software in 1997, which was acquired by IBM in 2003

1991 – LoadRunner was released by Mercury Interactive

- Acquired by HP in 2006 for $4.5B

Performance expertise moves to performance testing teams



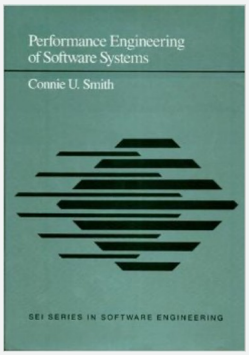
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1990: Software Performance Engineering

1990 – Dr. Connie Smith's book Performance Engineering of Software Systems was published

- 15 pages of bibliography

> Fix-it-later was once a viable approach, but it is now inappropriate and dangerous



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Does It Remind You Anything?

> Fix-it-later was a viable approach in the 1970s, but today the original promises no longer hold, and fix-it-later is archaic and dangerous. The original premises were:
>
> - Performance problems are rare.
> - Hardware is fast and inexpensive.
> - It's too expensive to build responsive software.
> - You can tune software later, if necessary.
>
> *Performance Engineering of Software Systems, 1990*

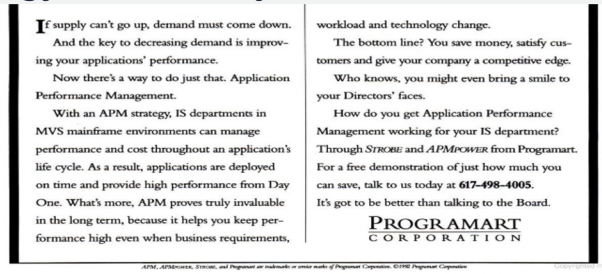
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1992: Application Performance Management (APM)

The term was coined by Programart Corp. (acquired by Compuware in 1999) in 1992 (in the mainframe context, as a combination of their STROBE and APMpower tools)

- STROBE had been on the market since the '70s

Introscope by Wily Technology, founded by Lew Cirne in 1998 (acquired by CA in 2006)



If supply can't go up, demand must come down.

And the key to decreasing demand is improving your applications' performance.

Now there's a way to do just that. Application Performance Management.

With an APM strategy, IS departments in MVS mainframe environments can manage performance and cost throughout an application's life cycle. As a result, applications are deployed on time and provide high performance from Day One. What's more, APM proves truly invaluable in the long term, because it helps you keep performance high even when business requirements, workload and technology change.

The bottom line? You save money, satisfy customers and give your company a competitive edge.

Who knows, you might even bring a smile to your Directors' faces.

How do you get Application Performance Management working for your IS department? Through *STROBE* and *APMPOWER* from Programart. For a free demonstration of just how much you can save, talk to us today at **617-498-4005**.

It's got to be better than talking to the Board.

PROGRAMART CORPORATION

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1998: End-User Monitoring

1998 – ETEWatch released by Candle

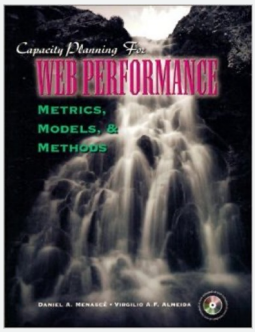
- Application response time monitor
- End-to-End Watch
- Candle acquired by IBM in 2004, then a part of Tivoli

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Mid 90s: Web

Most of existing expertise was still applicable to back end

Books applying existing expertise to Web (1998)





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2001 – Agile Development

The Manifesto for Agile Software Development is a foundational document written in February 2001 by 17 software practitioners at a ski resort in Snowbird, Utah

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

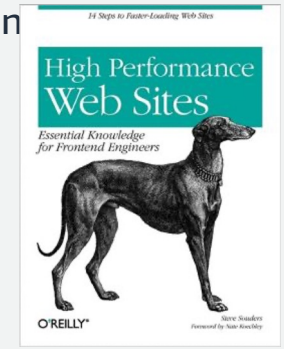
-Individuals and interactions over processes and tools
-Working software over comprehensive documentation
-Customer collaboration over contract negotiation
-Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

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2007: Web Performance Optimization

2007 – Steve Souders' book High Performance Web Sites: Essential Knowledge for Front-End Engin



New Web-specific expertise

2008 – WPO Community

- First Velocity conference
- WebPerf meetups

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2006: Cloud Computing

References to "cloud computing" appeared as early as 1996, with the first known mention in a Compaq internal document

Cloud computing was popularized with Amazon releasing its Elastic Compute Cloud product in 2006

Third-party autoscaling software Scalr and RightScale appeared in 2008. Amazon launched its own autoscaling feature in 2009.

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Peer-to-Peer, IoT, Fog, Edge [Decentralization]

Mark Weiser's 1991 paper on ubiquitous computing, "The Computer of the 21st Century”

The term "Internet of Things" (IoT) was coined by British technology pioneer Kevin Ashton in 1999

Akamai began using the specific term "edge computing" in 2002 for use of decentralized servers

The term "fog computing" was first introduced by Cisco in 2012

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Artificial Intelligence

1956: The field of AI research was founded at a workshop at Dartmouth College.

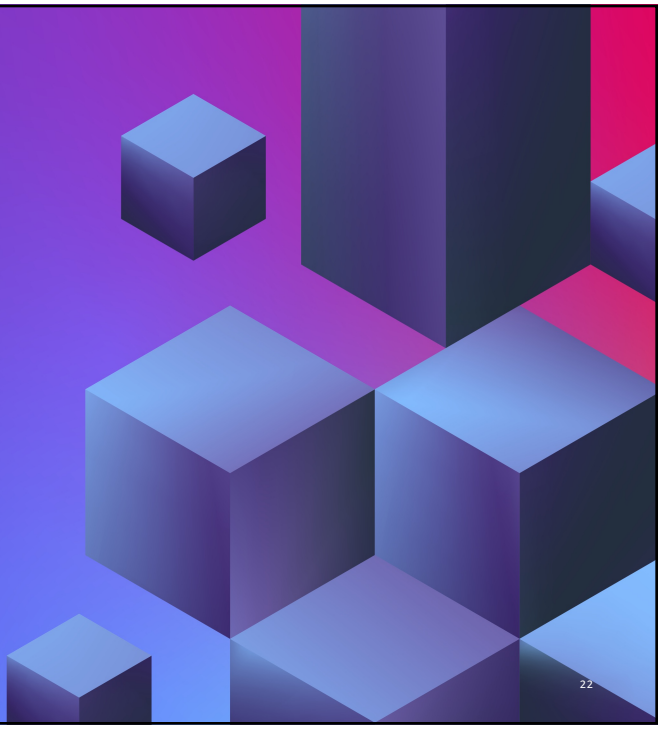
2017: Google researchers introduced the transformer architecture in their paper "Attention Is All You Need".

2018: GPT-1 and BERT launched, proving that pre-training neural networks on large text corpora could effectively capture general language context

2022: consumer-facing chatbot ChatGPT launched, bringing conversational LLMs into mainstream public awareness.

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Performance Gets More Attention



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Algorithmic Complexity

Time Complexity

Space Complexity

Big-O notation

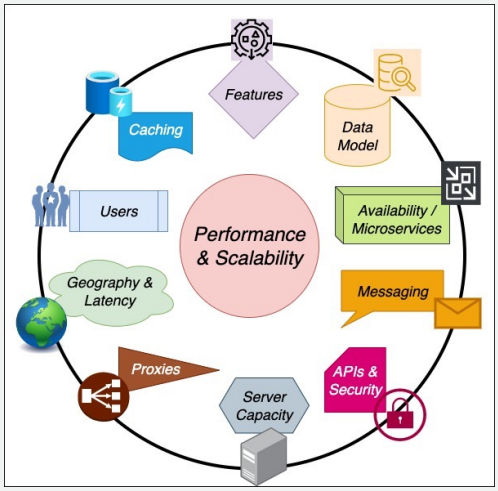
Almost in every interview around the globe !

Connect it with practical performance engineering?

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System Design Interview Cheat Sheet
by Vahid Dejwakh

Just ***an example*** of the changing attitude



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AWS Well-Architected Framework

The 6 Pillars of the AWS Well-Architected Framew

- Operational Excellence
- Security
- **Reliability**
- **Performance Efficiency**
- **Cost Optimization**
- Sustainability

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Well-Architected Trend Across the Industry

Azure Well-Architected Framework

Pillars: Performance Efficiency, Cost Optimization, etc.

Google Cloud Well-Architected Framework

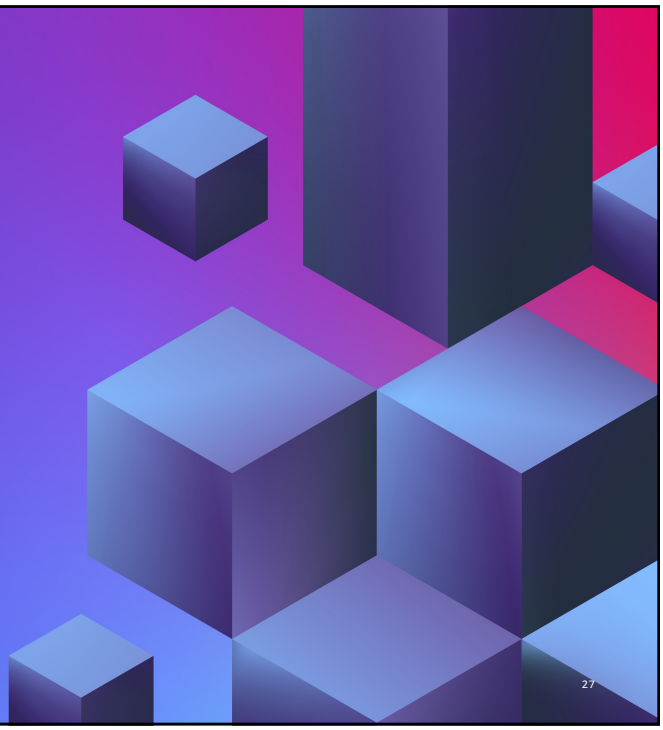
Pillars: Performance Optimization, Cost Optimization, etc.

Well-Architected Framework for Oracle Cloud Infrastructure

Pillars: Performance and Cost Optimization, etc.

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Adjusting Performance Engineering to Major Industry Trends



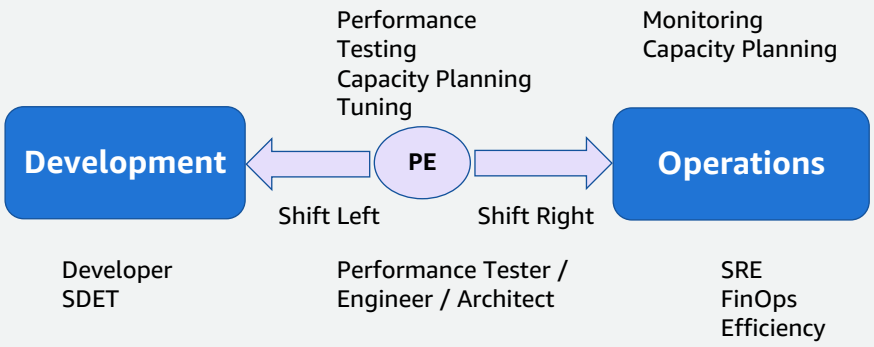
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What Changed?

- Web
  - Centralization, Open / unlimited workload
- Cloud
  - Further centralization [direct cost to performance correlation]
  - Dynamic configurations / Self-Management
- Agile / Iterative development
  - Continuous Integration / Delivery [fuzzier line between Dev and Ops]

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Integrating Performance Engineering into DevOps



Expand or be Squeezed Out ?

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Shift Left Integration with Development

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Adjusting to Agile and CI/CD

- Agile development changes the performance testing dynamics
  - Working system on each iteration from the beginning
  - You need a performance engineer for the whole project
    - ***Savings come from detecting problems early***
- Traditional Performance Engineering teams don't scale well
  - Increased volume (early testing, each iteration) exposes the problem
- Remedies: automation, making performance everyone's job
  - Shift Left
  - Self-Service

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Need for Continuous Performance Testing

- Integration into Agile and CI/CD
  - To catch regressions early
  - Collecting all info needed to investigate
  - From realistic testing to coverage, from SLO checking to differences between builds
- Foundation to build further automation on the top of it
  - For further performance optimization

Continuous Performance Testing



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Challenges of Continuous Performance Testing

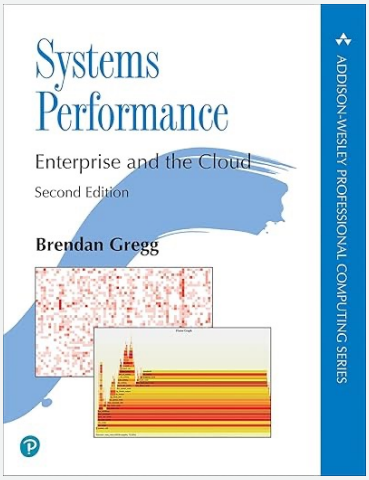
- Integration
- Coverage Optimization
- Variability / Noise Reduction
- Change Detection
- Advanced Analysis
- Operations / Maintenance
- Self-Service

Modern Challenges in Performance Testing



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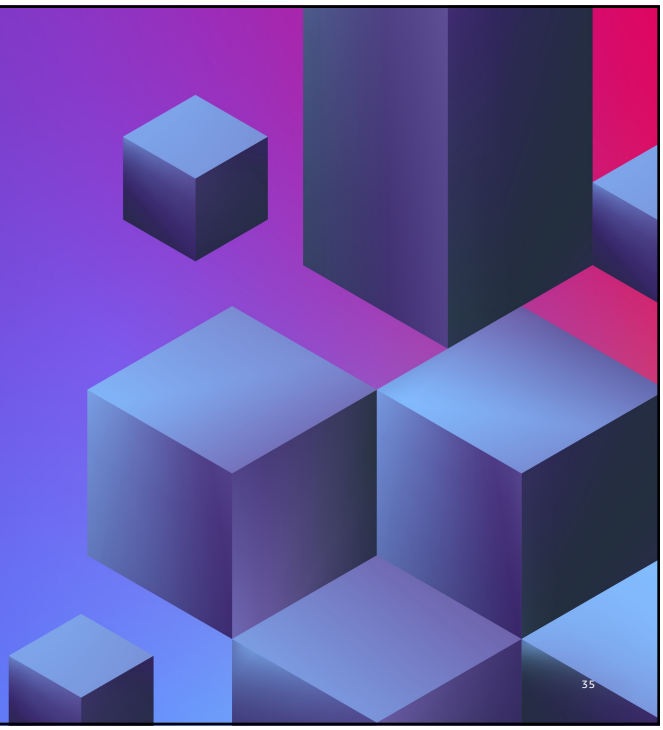
Systems Performance Book



- 886 pages
  - Modeling, 7 pages – including USL and Queueing theory
  - Large detailed sections on OSs, CPUs, Memory, File Systems, Disks, Network, perf, Ftrace, BPF
  - Chapter 12. Benchmarking – no mention of load testing tools

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Shift Right Integration with Operations



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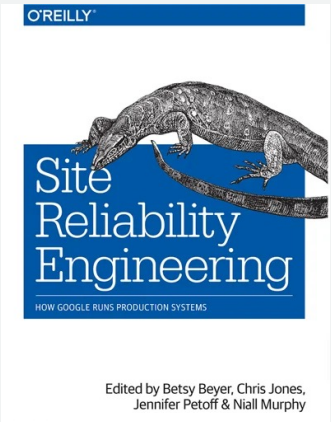
Observability

- Significant advances in tools (under observability umbrella)
  - Allow to do more PE work in production
    - In spite of more sophisticated environments
  - Three pillars: metrics, logs, traces
  - Further PE integration with operations

[Notion: Systems are too complex to properly test / reproduce problems]

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Site Reliability Engineering (SRE)



- The Book
- Tenets of SRE
  - "SRE team is responsible for the **availability, latency, performance, efficiency, change management, monitoring, emergency response, and capacity planning** of their service(s)."
  - None is considered separately in principles and practices
    - Except Service Level Objectives

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FinOps

- FinOps Foundation www.finops.org
  - Organization, framework, certifications, etc.
- Focus on cost
- The Foundation making drastic steps to expand
  - Cost-Aware Product Decisions
  - The Scope of FinOps Extends Beyond Public Cloud

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AWS Cloud Financial Management



The four pillars of Cloud Financial Management.

From Cost Optimization Pillar

- Cloud Financial Management with AWS

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Performance Engineering as an Independent Discipline

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Performance Engineering as an Independent Discipline

- Not well defined, no agreed Body of Knowledge, no certifications
- Fractured
  - Performance Testing
  - Web Performance
  - Capacity Planning
  - Etc.
- No significant developments recently [comparing to the increased attention]
  - Most developments are in integrated areas

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Decreased Attention - Queuing Models

- Multi-core servers, horizontal scalability, and not using servers to 100% probably allow using linear models in many cases
- "Black Box" ML models are not a replacement

The Forgotten Art of Performance Modeling



[Looks like we see a new wave of performance modeling for sophisticated AI systems]

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Decreased Attention - Load Testing Tools

- Performance engineering shifted to
  - Other ways to mitigate performance risk
    - As the results of centralization and more complex systems
  - More closely integrated continuous performance testing
    - Proprietary solutions
    - Part of application platforms (?)

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Failed Attempts to consolidate Performance Engineering

- Multiple attempts to consolidate Performance Engineering – all failed so far
  - HyPerformix certifications
  - CMG
  - DevPerfOps Foundation
  - and many others

- No organization, no body of knowledge, no certification [vendor – neutral]
  - There are vendor-specific certifications
  - ISTQB Certified Tester Performance Testing (CT-PT)



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Performance Titles Evolution

Used To Be

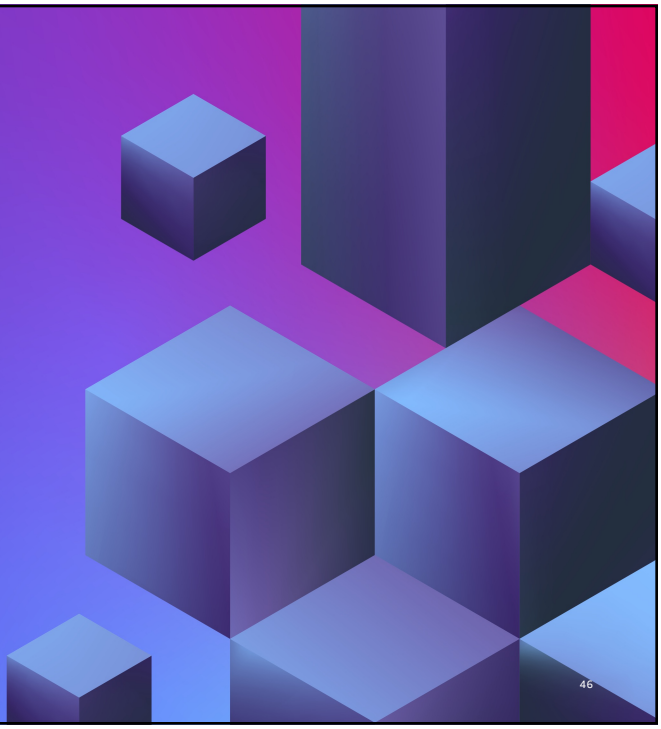
- Performance Tester
  - Create and run tests, simple analysis
  - More junior team member
- Performance Engineer
  - Same + deep analysis
- Performance Architect
  - Same + strategy

Now

- Performance Tester
  - Create and run tests, simple analysis
  - Mostly outsourced
- Performance Engineer
  - Somebody to make your software fast
- Performance Architect
  - Somebody to make your hardware fast

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AI Current Impact



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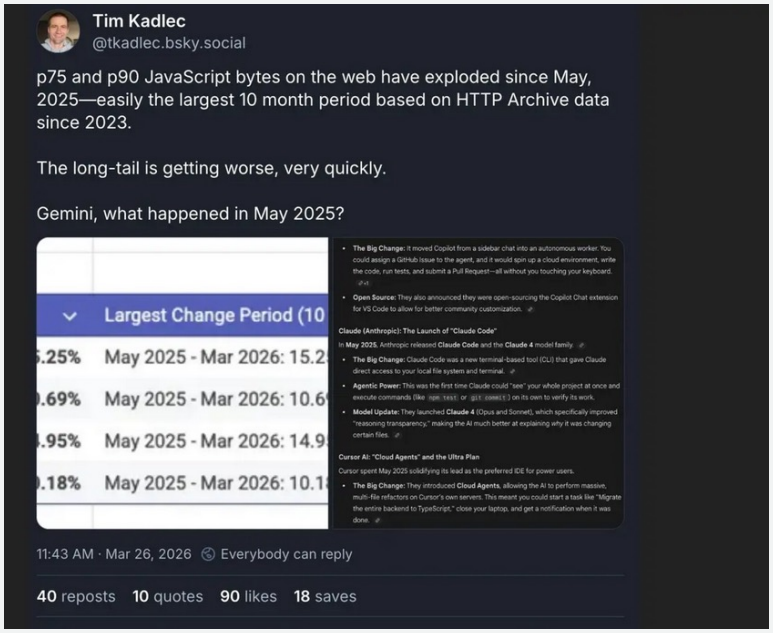
Current Impact

- Shows up everywhere, but not changing PE fundamentals
- Different aspects here, for example:
  - Systems Performance (Side effect of vibe coding)
  - Performance testing / optimization of AI-based systems
    - One more technology with its own challenges
  - Using AI in performance testing
  - Using AI in performance monitoring and tuning
  - Using AI in code optimization (existing or AI-generated)

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AI Side Effects

from Paul Calvano's Introduction to HTTP Archive



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Performance Testing / Optimization of AI-based systems

- The first challenge is that performance and benchmarking usually refer to correctness (response quality), not classic performance (throughput and latency).
  - Only way to separate is from context
- Some good information started to show up
  - AWS Well-Architected Framework - Machine Learning Lens
  - AWS Well-Architected Framework - Generative AI Lens





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AI-based Systems

- Many moving parts
  - Training
  - Inference
  - RAG
  - Agents
- Apparently one of challenges is keep the quality of the output during the test
  - Correctness as a constraint

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Using AI in Performance Testing

- Cases that appears to be promising for AI
  - Analysis
  - Reporting (Summarization)
  - Workload Modeling
  - Correlation (creating correlation rules)
- Concern: one of the main advantages of performance testing is reproducibility
  - AI output *DO* vary
    - We probably don't want it to mess up with the core load generation

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AI in Load Testing Tools

- Model Context Protocol (MCP) Servers provide a framework that connects AI models to the tool
  - Tricentis: Why MCP is a major leap in the evolution of AI-driven software testing
  - OpenText Performance Engineering: AI and MCP in Scripting, Analysis, and Agent Workflows
  - BlazeMeter: MCP server
- Top 5 AI Load Testing Tools in 2026

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Using AI in Performance Monitoring and Tuning

- Alerting and tuning looks promising for AI in case when we have a lot of data
  - Stable production systems
  - Many deployments (support)
  - Panda: Performance debugging for databases using LLM agents
  - Auto-WLM: Machine learning enhanced workload management in Amazon Redshift
- May not work well for new / modified systems

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Using AI in code optimization

- Quite a lot of research
- Some practical applications
  - AI Code Optimization in 2026: Tools, Tactics, and How to Roll It Out
  - Profile Guided AI Optimization with Zymtrace
- A good case for coding / standard performance optimization
  - May not work well for new / unique issues

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AI is Changing the Paradigm



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Spec-Driven Development

- Moving from ad-hoc prompting to structured, specification-based workflows that produce production-ready code
  - Kiro KIRO
  - Kiro and the future of AI spec-driven software development

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AI Will Generate Binary

Can AI Bypass Compilers? Evaluating the Feasibility of Prompt-to-Binary Systems

Elon Musk predicts that AI will bypass coding entirely by the end of 2026 - just creates the binary directly

AI Is Writing Code the Wrong Way by Waseem Alshikh

- AI generates binary, no high-level languages
  - No code-level changes, optimization
  - Performance [etc] need to be enforced / verified though context / framework

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Specification-Based Performance ?

- All Quality Attributes should be specification based
  - Performance, reliability, security, etc.
  - aka “Non-Functional Requirements”
- Performance Requirements: the Backbone of the Performance Engineering Process

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Summary

- Past: 60 years of fascinating history
- Present: Systems scale and sophistication skyrocket – performance gets more attention
- Future: AI is changing the development paradigm, details TBD
- The feeling that we are close to solving performance problems exists last 60 years
  - Probably will stay with us for a while

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